# Opreation Research:-

## (2017)

### (a)

Define OR and characteristics of OR?

Operation research (OR) is a multidisciplinary field that deal with the application of advanced Analytical method to help make better decisions and Solve complex problem in verious field including, business, engineering, economics, and healthcare

Charactriestics of OR:-

(i) Interdisciplinary:-

OR Combines Concepts from mathematics, statistics, computer, science, engineering and economatric

(ii) Analytical:-

OR uses qualitative methods to Analyze and model Complex System

(iii) model based:-

OR uses mathematical and Computational model to represent real world system.

(IV) Data driven:-

OR relies on data Analysis and interpretation.

(b)

Phases of solving a problem using OR:-

phase 1:- problem Definition

1:- Develop a clear problem statement

2:- define the objectives and goals

3:- Determine the scope and boundaries of the problem.

phase 2:- problem Analysis

1:- Breakdown the problem into smaller sub-problem

2:- identify key variables and relationships

3:- Develop a Conceptual model of the problem.

phase 3:- Collect Data

1:- Relevant data from various Sources

2:- format the data for Analysis

3:- Identify data and limitations:-

(C)

Summarize the steps for solving a Lp model?

step1:- formulate the problem

1:- Define decision variable

2:- Determine objective function

3:- Write the Lp model in mathematical from

step2:- ghrafical representation

1:- plot the graph

2:- Identify the region

step3:- Convert the standered from

1:- Inderduce variables

2:- Write Lp model in standered from

Step4:- Choose a solution method

1:- Graphical method

2:- Simplex method

3:- Dual simplex method

4:- Interior Point method

(d)

Define the following with reference to LLP

(i) unbounded Solution

(ii) Alternative optima

(i) Unbounded Solution:-

- There is no limits to how good your answer can be
- you can keep improving your Solution forever
- These are no constraints to stop you

Example:-

Imagine you can make unlimited money with no restrictions

(ii) alternative optima (multi best Solution)

- There is more than one best Solution
- All Best Solutions give the Same results.
- You can chose any of these Best solution

Example:-

Imagine you have two different routes to work that make the same amout of time.

(c)

How the problem of minimization can be solved linear programming

Same answer question No: (c)

# (2018)

## (a)

**Objective function:-**

In Operations research (op the objective function is a expression that represents the goal or objective linear programming (LP) problem.

*minimization problem:-*

$$\text{minimize } Z = C_1X_1 + C_2X_2 + \dots + C_nX_n$$

*maximization problem:-*

$$\text{maximize } Z = C_1X_1 + C_2X_2 + \dots C_nX_n$$

Where

- $Z$ is the objective function value
- $X_1, X_2 \dots X_n$ are the decision variables
- $C_1, C_2 \dots C_n$ are the Coefficient of the decision value

**Type of objective function**

*linear objective function:-*

A linear

Example:-

$$\text{maximize } z = 2x_1 + 3x_2$$

2:- Non-linear objective function:-

A non-linear combination of decision variables.

Example:-

$$\text{maximize } z = 2x_1 \hat{} 2 + 3x_2$$

Decision variable:-

A Decision variable is a variable that represents a decision or a choice that needs to be made in a problem.

characteristics:-

1:- Controllable:- Decision variable are under the control of the decision-maker.

2:- Quantifiable:- Decision variable can be measured and quantifiable.

3:- Variable:- Decision variable can take on different values.

Type of Decision Variable:-

1:- Continuous Decision Variable

2:- Integer Decision Variable

3:- Binary Decision Variable

4:- Catagorical Decision Variable

(C)

UnContrallable Variable:-

An uncontrallable variable is a variable that is outside the Control of the decision-maker and whose value is determined by external fector.

Charactriestics:-

1:- External influence:- UnControllable variable are influenced by external fector, such as market trends weather, or goverment palicies

2:- No Control:- Decision-makers have no Contral over the value of uncontrolled variable

3:- fixed value:- UnControllable variable have a fixed value that is determined by external factors

(d)

Constraints:-

In Opreations Research (OR), Constraints are limitations or restrictions on the value of decision variable Constraints define the feasible region of the problem, which is the set of all possible solutions that satisfy all the Constraints

Types of Constraints:-

1 Equality Constraints
2 inequality Constraints
3 Non-negativity Constraints
4 integer Constraints
5 Binary Constraints

(e)

Deterministic model:-

In operations Research a deterministic model is a mathematical representation of a system or problem where all the parameters and variables are known with Certainty.

## Characteristics of Deterministic model

(1) known parameters:- all parameters and coefficients are known with Certainty

(2) No uncertainty:- There is no uncertainty or randomness in the model

(3) unique Solution:- The model has a unique Solution that can be determined using mathematical technique.

## Type of deterministic model

(1) linear programming
(2) integer programming
(3) Dynamic programming

# (2019)

## (a)

Define OR and briefly discuss its impact on society.

Define OR → Repeat

Impact of OR on Society:-

1) improved Efficiency:- OR has helped optimize processes reducing waste and increasing productivity in various industries.
2) Better decision-making:- OR provides decision-makers with powerful tools to Analyze Complex problems and make informed decisions.
3) Enhanced Customer Experience:- OR has improved customer service in industries like healthcare, finance, and transportations.

## (b)

phases of solving a problem by using OR:

Repeat. 2017 (b)

## (C)

Explain the steps involved in setting up of a simplex method

Simplex Method in easy to understand terms:

(1) write the problem:- Define the problem and write it in a mathematical from.

(2) Make it standered:- Convert the problem into a standered from

(3) Create a table:- make a table with all the information

(4) choose a pivot:- Selet a number to choose a povit

(5) pivot a table:- use the pivot to change the table

(6) Repeat:- keep reapeating steps 4-5 until you get the answer.

(7) Read the answer:- look at the final table and read the solution.

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## (d)

Define the following with reference to LPP
(I) feasible Solution
(II) Slack variable

(I) feasible Solution:-

A feasible variable is a decision variable that satisfies all the constraints of the LPP

Example:- You have 10 hours to finish a project. The number of hours you work (x) is a feasible variable if you work between 0-10 hours

(ii) Slack variable:-

A slack variable is a variable introduced to convert an inequality constraint it represent the amount of unused resource or the different between the actual and maximum value of a constraint

Example:-

You have 10 hours to finish a project but you only work 8 hours. The slack variable (s) is a 2 hours which is unused time.

How it is ensured that variable are positive in graphical method

In the graphical method, variables are ensured to be positive by:

(1) Non-Negativity Constraint:- Adding Constraints to the problem such as $x \geq 0$, $y \geq 0$ to ensure that the variable are non-negative

(2) graphing on the first Quadrant:- graphing the constraints and the objective function only in the first quadrant (where $x \geq 0$ and $y \geq 0$) to ensure that the Solution is positive

## (2021)

## (a)

Define Opreation research and Discuss its role in decision making?

Repeat → OR → 2017, 2019

Discuss its role in decision making:-

(1) improved Decision Quality:- OR helps to make batter decision by analyzing complex systems and evaluating alternatives

(2) Increase efficieny:- OR helps to optimize resources and improve productivity

(3) Reduced Risk:- OR provides tools and techniques for managing risk and uncertainty.

## (b)

Discuss the methods used to Solve OR models?

1) linear programming
2) integar programming
3) Dynamic programming
4) Queuing theory
5) Simulation
6) Heuristics
7) Metaheuristics

## (C)

Define linear programming and discuss its properties?

Linear programming (LP) is method used to optimize a linear objective function, Subject to a set of linear constraint. it is a powerful tools used to make decision in various field, such as business economics and engineering.

## properties of Linear programming

(1) linearity:- The objective function and all certaint must be linear.
(2) Non-negativity:- All decision variables must be non-negative
(3) Divisibility:- Decision variables can takes on any value, including fractions and decimals
(4) Certainty:- all parametars and Coefficient are known with Certainty.

## Define graphical Sensitivity analysis?

graphical sensitivity Analysis is a technique used to Analyze how changes in the Coefficients of a linear programming (LP) problem affect the optimal solution, visually represented through graphs

Define Simplex method. Discuss its main feature and Conditions?

The Simplex method is a popular algorithm for ~~linear~~ Solving linear programming (LP) problems.

Main features of simplex method

(1) Iterative process:-

The simplex method involves a series of iterations to find the optimal solution.

(2) pivot Element:-

A povit element is choosen to move from one basic feasible solution to another.

Condition for applying simplex method

(i) linear objective function:- The objective function must be linear

(ii) linear Constraint:- All Constraint must be linear.

(f)

Discuss the iterative nature of the simplex method.

The simplex method is an iterative process that involves a series of step to find the optimal Salution.

(i) Chacking for optimality
(ii) Seleciting the entering variable
(iii) selecting the leaving variable
(iv) pivoting
(v) Repeating

## (2022)

### (i)

Dealing with unrestricted variable?

An unrestricted variable is a variable that can take on any value positive, negative, or zero, without any restrictions.

Example:-

maximize : $2x + 3y$

subject to: $x + 2y \leq 4$

### (ii)

phases of operation Research study?

Repeat 2017 (b) 2019 (b)

### (iii)

What are steps in graphical soultion of problem?

1. Define the Decision variable
2. plot the constraints
3. Identify the feasible region
4. plot the objective function
5. Determine the optimal Solution
6. Read the optimal Solution
7. check for multiple optimal Solution

(iv)

What is stochastic models give two example

Stochastic models:-
Stochastic model are math tools that help us understand things that involve chance or uncertainty.

(1) Queuing model: A queuing model is a stochastic model that is used to analyze the behaviour of customers waiting in line for service

(2) Stock price model: A stock price model is a stochastic model that is used to predict the future price of a stock.

(v)

How it is ensured that variables are positive in graphical method

prepeat 2019 (e)

(2023)

(i)

Slack variable:-

Repeat 2021 (d)

(ii)

Constraints

Repeat 2018 (d)

(iii)

Optimal Solution:-

An optimal solution is the best possible solution to a problem given certain constraints and objectives find optimal Solution:-

1 Define the problem
2 formulate a ~~pb~~ Model
3 Solve the model
4 Analyze the results

(iv)

Degeneracy:-

Degeneracy is a problem that happens when we are trying to solve a linear programming problem. it's like getting stuck in a loop

(V)

Alternative optima:-

alternative optima accur when a linear programming problem has multiple optimal Solution:-

Method to find AO
1 Simplex method
2 Dual simplex method